THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 15 **(b)** 17 **(c)** 14 **(d)** 16

2. 2^7 added 2 times equals: 1

(a) 2^8 **(b)** 4^7 **(c)** 2^{14} **(d)** 2^9

3. If $(34)_b = 19$ in decimal (a two-digit number in base b), then b equals: 1

(a) 4 **(b)** 6 **(c)** 7 **(d)** 5

4. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 198° **(b)** 90° **(c)** 102° **(d)** 144°

5. The smallest number which leaves remainder 5 when divided by 12 and 18 is: 1

(a) 41 **(b)** 221 **(c)** 36 **(d)** 77

6. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 61 **(b)** 64 **(c)** 63 **(d)** 62

7. If $2^{2x+1} = 2^9$, the value of x is: 1

(a) 8 **(b)** 4 **(c)** 5 **(d)** 3

8. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 31 kg is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

9. One angle of a parallelogram is 2 times its adjacent angle. The smaller angle is: 1

(a) 120° **(b)** 80° **(c)** 60° **(d)** 90°

10. The number of multiples of 7 between 100 and 500 (both included) is: 1

(a) 57 **(b)** 58 **(c)** 56 **(d)** 59

11. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 80 **(b)** 77 **(c)** 79 **(d)** 78

12. The number of digits in $2^4 \times 5^7$ is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

13. **Assertion (A):** The number 1296 is a perfect square. 1
Reason (R): A number is a perfect square if every prime in its prime factorisation occurs an even number of times.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** $2^{-3} = \frac{1}{8}$. 1
Reason (R): For a non-zero number a, $a^{-n} = \frac{1}{a^n}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The binary number 101_2 equals 5 in decimal. 1
Reason (R): In base 2, the value of a number is the sum of each digit times a power of 2, with the rightmost place being 2^0.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the smallest number to be added to 3900 to make it a perfect square. What is the square root of the result? 2

17. Find x if $3^{2x-1} = 243$. 2

18. Find the base b if $(23)_b = 13$ in decimal. 2

19. In a parallelogram ABCD, $\angle A = (4x + 14)^\circ$ and $\angle C = (5x + 2)^\circ$. Find x and $\angle B$. 2

20. Find the smallest number which leaves remainder 3 when divided by 20 and 25. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A square garden has area 1225 m². Fencing costs ₹40 per metre. Find the total cost of fencing it. 3

22. A sheet of paper 0.1 mm thick is folded in half 10 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

23. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 45 g? Explain using binary numbers. 3

24. In a parallelogram ABCD with $\angle A = 50^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

OR

Show that the six-digit number 247247 is divisible by 7, 11 and 13. 3

25. A solid cube of side 6 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

Light travels 3×10^8 m in one second. How far does it travel in 2 minutes? Give the answer in scientific notation. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ramesh, a gardener in Jaipur, is given a square plot of area 1764 m^2 to plant a lawn and fence it. He wants to find its dimensions using prime factorisation. 5

(i) Express 1764 as a product of prime factors.

(ii) Find the side of the plot.

(iii) Find the cost of fencing all four sides at ₹50 per metre.

(iv) The owner decides to increase the side to 45 m. Find the extra area added.

27. In some base b the number $(234)_b$ equals 69 in decimal. Find b and verify your answer by expanding. 5

OR

In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 1 : 2 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

END OF QUESTION PAPER Total: 50 Marks

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